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Exercise 1: Comparative study of low-code development versus traditional application development approaches.

Aim:

The aim of this study is to compare **Low-Code Development** and **Traditional Application Development** approaches in terms of development process, coding requirements, development speed, cost, flexibility, maintenance, and overall efficiency. The study helps organizations choose the most suitable development method based on their business needs.

Introduction to Traditional Application Development:

Traditional Application Development is the conventional software development approach where developers write source code manually using programming languages such as Java, Python, C#, JavaScript, and others.

In this approach, every component of the application—including user interfaces, business logic, database interactions, and integrations—is developed through coding. The Software Development Life Cycle (SDLC) is followed, consisting of planning, design, coding, testing, deployment, and maintenance.

Characteristics:

- Requires skilled programmers.
- High level of customization and flexibility.
- Longer development time.
- Suitable for complex enterprise applications.
- Higher maintenance effort.

Examples:

- Banking systems
- E-commerce platforms
- Enterprise Resource Planning (ERP) systems
- Healthcare management systems

Introduction to Low-Code Development:

Low-Code Development is a modern software development approach that minimizes manual coding through visual interfaces, drag-and-drop components, pre-built templates, and automated workflows.

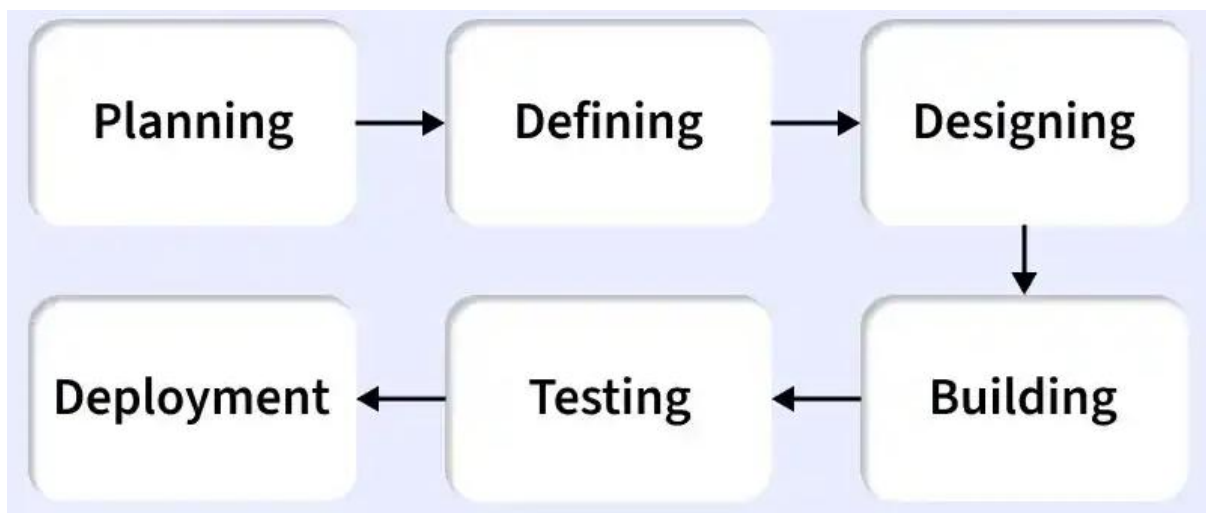
Low-code platforms enable both professional developers and non-technical users (citizen developers) to create applications rapidly with minimal programming knowledge.

Characteristics:

- Visual development environment.
- Faster application delivery.
- Reduced coding effort.
- Lower development costs.
- Easier maintenance and updates.

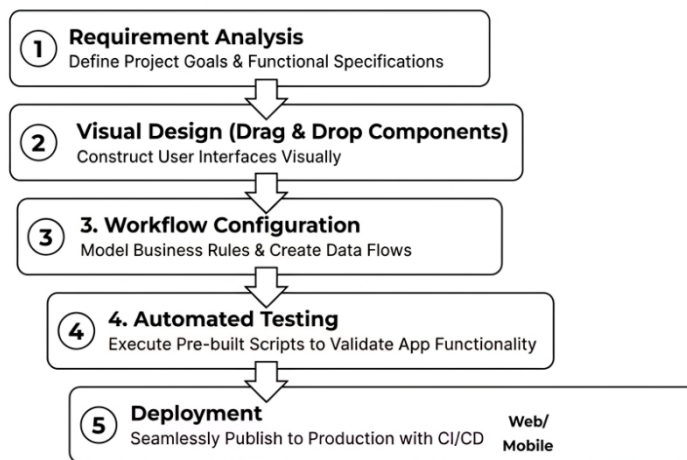
Examples of Low-Code Platforms:

- UiPath
- Microsoft Power Apps
- OutSystems
- Mendix

Diagram: Stages in Traditional and Low-Code Development:**Traditional Development Process:****Description:**

- Coding is the most time-consuming phase.
- Multiple teams may be involved.
- Changes require code modification and retesting.

Low-Code Development Process:



Description:

- Minimal manual coding.
- Pre-built components reduce development effort.

Comparison between Traditional and Low-Code Development:

Parameter	Traditional Development	Low-Code Development
Coding Requirement	Extensive manual coding	Minimal coding
Development Speed	Slower	Faster
Technical Skills Needed	High	Moderate to Low
Development Cost	Higher	Lower
Customization	Very High	Limited to Platform Capabilities
Scalability	Highly Scalable	Platform Dependent
Maintenance	More Complex	Easier
Testing Effort	Manual and Time-Consuming	Mostly Automated
Deployment Time	Longer	Shorter
Target Users	Professional Developers	Developers and Citizen Developers
Integration	Requires Coding	Built-in Connectors Available
Best Suited For	Complex Enterprise Applications	Rapid Business Applications

Result:

The comparative study shows that low-code development enables faster application development with less coding effort, while traditional development provides greater flexibility and customization.